BTS(C) – VI – 10 – 007 – E

B. Tech Degree VI Semester Examination, April 2010

ME 603 INSTRUMENTATION THEORY AND CONTROL ENGINEERING

(2002 Scheme)

Time : 3 Hours — Maximum Marks : 100

I. (a) Develop a mathematical model for a second order system. Discuss on its step response with the help of relevant plots and equations. (8)

(b) Discuss the various types of errors . A 600V voltmeter is specified to be accurate within $\pm 2\%$ off full scale. Calculate the limiting error when the instrument is used to measure 250V. (7)

(c) Describe the following terms.

(i) loading effect (ii) backlash
(iii) dead space (iv) static sensitivity
(v) threshold (5)

OR

II. (a) With the help of a neat block diagram, explain the functional elements of a measurement system with pressure thermometer as an example. (7)

(b) Explain seismic instrument as a second order instrument. (6)

(c) A liquid thermometer has a glass bulb protected by a well. The system can be represented by a double capacity system with time constants of 40S for the well and 20S for the bulb. The thermometer is subjected to a cyclic change of $\pm 10^0 C$ which occurs every 120S. Find the maximum value of the indicated temperature and delay time. (7)

III. (a) What are strain rosettes? What do you mean by temperature compensation in strain gauge. (7)

(b) With the help of neat sketches explain the working of hydraulic and pneumatic load cells. (6)

(c) Explain –

(i) Heenan and Froude hydraulic dynamometer
(ii) Beam and strain gauge transmission dynamometer. (7)

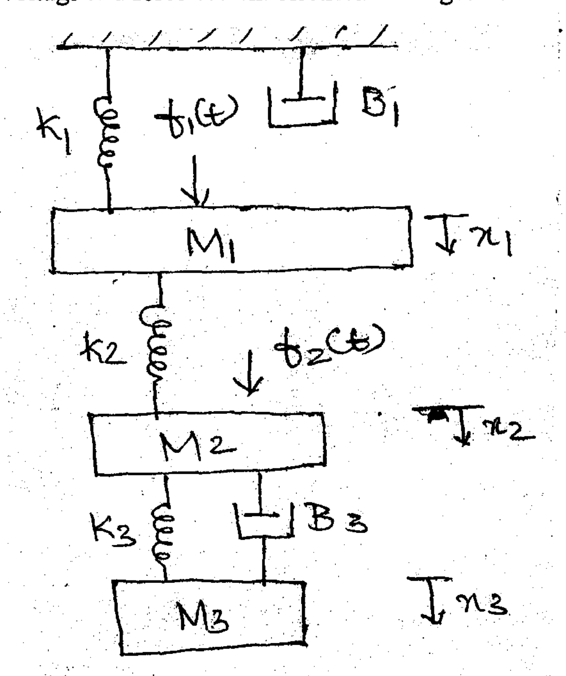
OR

IV. (a) Write notes on optical pyrometers and total radiation pyrometers. (7)

(b) Explain (i) Infrared pyrometry
(ii) Gas Chromatography (7)

(c) Explain the working principle of
(i) Sound level meter
(ii) Geiger Muller Counter. (6)

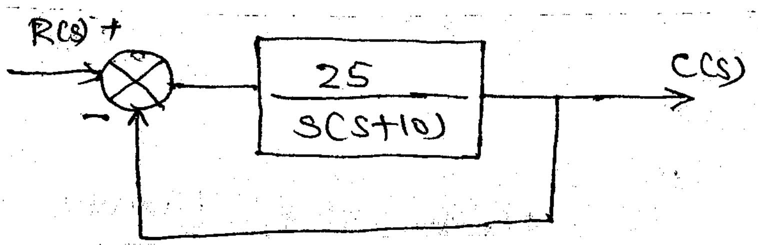
V. (a) Write the differential equations governing the mechanical system shown in the figure. Draw force voltage and force current electrical analogous circuits.



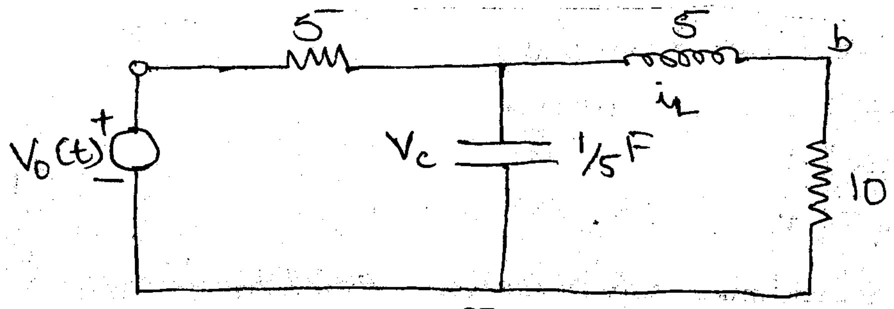
(7)

(Turn Over)

(b) Derive the transfer function of the closed loop system given below. (6)



(c) Write the state equation in matrix form for the circuit given below. (7)



OR

VI. (a) The closed loop transfer function of a system is given by $\frac{C(s)}{R(s)} = \frac{7}{s^2+4s+5}$.

Determine. (i) the damping ratio (ii) natural frequency (iii) step response (iv) error response (6)

(b) Evaluate the dynamic error co-efficients for the unity feed back system with forward path transfer function. $G(s) = \frac{15}{s(s+2)}$. (7)

(c) The open loop transfer function of a unity feed back control system is given by $G(s) = \frac{0.4s+1}{s(s+0.6)}$. Determine the transient response for unit step input and sketch the response. Evaluate the maximum overshoot and the corresponding peak time. (7)

VII. (a) What is Laiponov function? How it is being developed. Explain how it can be used in the stability analysis of the system. (10)

(b) Using Routh criterion investigate the stability of a unity feed back control system whose open loop transfer function is given by $G(s) = \frac{e^{-ST}}{s(s+2)}$. (10)

OR

VIII. (a) Draw the Bode plot for the transfer function $G(s)\frac{50}{s(1+0.25s)(1+0.1s)}$.

From the graph determine
- (i) Gain cross over frequency
- (ii) Phase cross over frequency
- (iii) G.M. and P.M
- (iv) Stability of the system. (12)

(b) Explain Absolute and relative stability. (8)

IX. (a) Describe the transfer function of AC servo motor and armature controlled DC servomotor and explain. (7)

(b) Explain the operating principle, application, types and characteristics of stepper motor. (7)

(c) Explain the construction, working principle and applications of synchros. (6)

OR

X. (a) Write notes on :
- (i) Pneumatic controllers (ii) Hydraulic controller
- (iii) Integral control action (iv) Derivative control action (8)

(b) Write notes on tachogenerators. (5)

(c) What are the advantages and limitations of derivative control action compared to others. (7)

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